

Self-Dual BKZ on NTRU: Dimension-Onset, Fatigue, and a Cross-Engine Reference-Free DSD Observable

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Abstract

This technical report extends the self-dual BKZ (SD-BKZ) reduction benchmark of the companion paper from LWE-Kannan lattices to NTRU , and from a single enumeration engine (fplll) to a second, sieving engine (G6K). We report three empirical results. First, on circulant NTRU lattices the SD-BKZ advantage exhibits a sharp *dimension-onset*: tied with BKZ for small n , a high-variance spike around $n \approx 71\text{--}73$ (mean advantage up to $+9.7$ nats), then a tight $+1.5$ -nat plateau for $n \geq 79$. Second, the defensible NTRU observable is *reference-free*: signed $d(\text{LN})$ is a reference artifact on NTRU (no canonical fixed point), but the BKZ-vs-SD-BKZ profile divergence / dense-sublattice-discovery (DSD) onset is robust, and the SD-vs-BKZ DSD-onset gap grows with dimension ($0\% \rightarrow 27\%$ over $n = 67\text{--}113$) as q crosses the fatigue point. Third, the cross-engine study confirms the companion paper’s thesis from the other side—the Root Hermite Factor is blind to the SD-BKZ difference while $d(\text{LN})/\text{DSD}$ sees it—now on a sieve oracle, and finds the effect to be regime-dependent. We also re-validate, on a new modulus family, an fplll GSO numerical fix. All experiments are reproducible via a containerised, determinism-gated benchmark.

Keywords: lattice reduction, BKZ, SD-BKZ, NTRU, overstretched, fatigue, dense sublattice discovery, Rankin profile, sieving, G6K

1 Introduction

The companion paper [CB26] showed, on LWE-Kannan lattices, that the Root Hermite Factor (RHF) is structurally blind to differences between BKZ and self-dual BKZ [MW16] that the Li–Nguyen Rankin profile distance $d(\text{LN})$ [LN20] makes visible. This report asks two follow-up questions a data paper is well suited to answer: (i) *does the SD-BKZ phenomenology generalise beyond LWE to NTRU?* and (ii) *does it reproduce on a fundamentally different reduction oracle—sieving (G6K) rather than enumeration?*

Our contributions are primarily empirical and infrastructural:

- A characterisation of the SD-BKZ advantage on circulant NTRU lattices [DvW21], including a sharp dimension-onset transition (Section 4).
- Evidence that the appropriate NTRU observable is reference-free, and a *dense-sublattice-discovery (DSD) onset* whose SD-vs-BKZ gap grows with dimension as q approaches the fatigue point (Section 5).

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- A cross-engine study (Section 6): G6K wired as a second engine behind a reproducibility-gated seam, a self-dual pump-and-jump construction validated against fpIII’s SD-BKZ, and confirmation that RHF-blindness is engine-independent but regime-dependent—consistent with independent observations of RHF-equality under a non-exact oracle [Row24].
- An independent re-validation of an fpIII GSO catastrophic-cancellation fix on a new (NTRU) modulus family (Section 7).

This is a *technical report / data paper*: the emphasis is on the experimental apparatus, the data, and reproducibility, rather than on new theory. Closest prior work is Deng and Jia’s self-dual BKZ over a sieve oracle [J⁺23]; we differ in measuring the SD-vs-BKZ structural difference via the reference-free profile observable rather than swapping the oracle for speed. White [Whi26] studies dynamic block-size strategies within the same Li–Nguyen framework, complementary to the fixed- β , q -swept regime characterised here.

2 Background

2.1 Circulant NTRU lattices and fatigue

We use the circulant NTRU construction of Ducas and van Woerden [DvW21]. For a prime n and modulus q , sample ternary secrets $f, g \in \mathbb{Z}_q[x]/(x^n - 1)$ (uniform $\{-1, 0, 1\}$, coordinate variance $\sigma^2 = 2/3$) with f invertible, set $h = g/f \bmod q$, and form the $2n \times 2n$ basis

$$B = \begin{pmatrix} qI_n & 0 \\ H & I_n \end{pmatrix},$$

where H is the circulant matrix of h . The secret (f, g) is an unusually short vector in the row span; recovering it breaks the key. Two structures distinguish B from a random q -ary lattice: a q -vector *head* (rows of qI_n that survive reduction as length- q vectors) and a *dense secret sublattice* spanned by the rotations of (f, g) . A BKZ-reduced NTRU profile therefore follows the *ZGSA* (Z-shaped) model—a flat head at $\log q$, then a GSA slope—rather than the single-slope GSA line of a random q -ary lattice.

As q grows at fixed n , the secret sublattice becomes relatively denser and easier to detect: at the *fatigue point* $q_{\text{fat}}(n) \approx 0.004 n^{2.484}$ [DvW21] the attack transitions from the standard regime (secret found only through full SVP) to the *overstretched* regime, where reduction discovers the dense sublattice (*dense-sublattice discovery*, DSD) well before solving SVP. The geometry of this transition—and how self-dual reduction moves through it—is the object of this report.

2.2 BKZ, SD-BKZ, and the two engines

Block Korkine–Zolotarev (BKZ) reduction repeatedly calls an SVP oracle on length- β projected sublattice windows, sweeping left to right. Self-dual BKZ (SD-BKZ) [MW16] alternates a forward (primal) pass with a backward pass on the reversed dual basis, so each tour reduces the basis from both ends. On LWE–Kannan lattices the companion paper [CB26] found the two produce bases of indistinguishable Root Hermite Factor yet measurably different Rankin profiles; this report asks how that difference behaves on NTRU and across two oracle implementations.

We use two engines. **fpIII** supplies an exact enumeration SVP oracle; its self-dual variant (**SD_VARIANT**) is our ground truth. **G6K** [ADH⁺19] supplies an approximate *sieving* oracle through the pump-and-jump BKZ tour (**pump_n_jump_bkz_tour**); we realise SD-BKZ on it as a primal pump-and-jump pass followed by a dual pass run under G6K’s **dual_mode** (operations on the dual basis, bounds reflected about $\text{full_n}/2$). The two

oracles differ fundamentally—deterministic enumeration versus randomised sieving—which is what makes a phenomenon that survives both engines credible. Sieving carries a fixed setup cost that makes small β uninformative, so all cross-engine experiments use $\beta = 40$, the smallest block size at which the sieve does non-trivial work; this is also the regime in which related q -ary forgery attacks operate [DEP23].

2.3 The reference-free DSD observable

The companion paper’s metric is $d(\text{LN}) = \frac{1}{m} \sum_i |r_i - \rho_i|$, where r_i is the slope-removed Gram-Schmidt log-norm profile and ρ_i is the Li-Nguyen BKZ fixed point [LN20]—a GSA line valid for a *random* q -ary lattice. NTRU is not random q -ary (Section 2.1): its reduced profile follows the ZGSA Z-shape, for which no closed-form Li-Nguyen fixed point exists. Comparing an NTRU profile to the LWE line therefore conflates structure with reduction dynamics, and the *signed* advantage $d(\text{LN})_{\text{BKZ}} - d(\text{LN})_{\text{SD}}$ can flip sign under a different reference—it is a *reference artifact*.

Two observables survive this. First, the *unsigned* profile divergence $\frac{1}{m} \sum_i |r_i^{\text{BKZ}} - r_i^{\text{SD}}|$ is reference-free (it never names ρ); on NTRU it is large precisely at the transition and reproduces under three independent references. Second, and what we report throughout, is a binary *dense-sublattice-discovery* flag: a reduction has found the secret sublattice when $b_1 = \min_i \log \|\mathbf{b}_i^*\| > 1.5$, where the threshold is the q -independent ternary-secret log-norm floor $\log \sqrt{2n \cdot 2/3} \approx 1.96$ (we also track the count of GS vectors that have collapsed below the q -vector head, $\leq n + 1$). Both are properties of the output basis alone, independent of any modelled reference. The onset q is the smallest modulus at which a given reduction flags DSD; the SD-vs-BKZ onset *gap* is the object of Sections 5 and 6.

3 Experimental Setup

3.1 Generators and engines

NTRU bases are built by `generators/ntru.py` following Section 2.1; every basis is checked structurally at construction ($\dim = 2n$, a qI_n block, and $Hf \equiv g \pmod{q}$) before any reduction. Both engines sit behind a single seam: `run_single` takes a `backend` argument, and `make_backend` returns either an `fpdll` or a G6K backend exposing a common `tour()/gso()` interface. The per-tour reduction loop is then engine-blind—the same loop drives enumeration and sieving, so the two cannot diverge through duplicated control flow. The G6K self-dual tour is a primal `pump_n_jump_bkz_tour` followed by a dual pass executed inside a `temp_params` block that flips `dual_mode`; this was validated against `fpdll`’s `SD_VARIANT` by checking that the SD-BKZ profile-change signature (the GS-slope flattening) agrees in direction and magnitude across engines (`fpdll` -0.00039 vs G6K -0.00040 per tour at $n = 80$, $\beta = 40$).

3.2 Reproducibility and determinism gates

Both engines run in containers built from a digest-pinned base with `snapshot.debian.org` apt pinning and a fixed `-march=x86-64-v3` target, so the binaries are bit-reproducible across machines. Each result is a per-seed JSON carrying the full input parameters and output profile; a SHA-256 manifest gates byte-identity. The `fpdll` path is held to exact-JSON reproducibility by `verify.sh`; the G6K path by `verify_g6k.sh` against a captured reference hash, confirmed on a second machine (the CI runner).

Determinism on the sieve required care (ADRs 005–008). G6K’s multi-threaded sieve has a nondeterministic reduction race, so all sieving runs single-threaded (`threads = 1`); across-seed parallelism still uses all cores, since seeds are independent. The `fpdll` global RNG must be seeded both before basis construction and again before the sieve (G6K

Table 1: SD-BKZ advantage on NTRU: tied $\rightarrow$ onset spike $\rightarrow$ plateau.

n	mean adv	std	seeds > 0
59–67	~ 0	tight	3–9/20
71	+4.35	5.59	16/20
73	+9.68	5.21	19/20
79–89	+1.5	~ 0.2	20/20

samples from it), and the `SieverParams` seed is a no-op that we never rely on. With these, multi-tour G6K runs are bit-identical run-to-run, and—verified at $n = 80$, $\beta = 60$, three tours—per-tour re-seeding is itself a no-op: the construction seed alone determines all N tours, so the policy is to seed once at construction. The new modulus and engine introduced no schema change to the seed JSON, preserving the companion paper’s SHA baselines.

3.3 Campaigns and parameter grids

Every experiment is a named campaign in `config/sweep.toml`, fixing generator, engine, n -grid, β , q , tour count, and seed count so a single command reproduces a cell. The dimension-onset sweep (Section 4) fixes $q = 97$, $\beta = 20$, 50 tours and sweeps $n \in [59, 89]$ at 20 seeds each. The fatigue and DSD-onset sweeps (Section 5) fix n and sweep q across the fatigue point. The cross-engine sweeps (Section 6) fix $n = 89$, $\beta = 40$, 50 tours and sweep a matched q -grid on *both* engines—the G6K campaign (`ntru_g6k_qsweep`) and the fplll campaign (`ntru_xeng_qsweep`) write to leaves that share the same (q , precision, tours, n , β) key, so a comparator reads the two engines cell-for-cell. The Kahan re-validation (Section 7) runs $n = 127$ at $q \in \{971, 1087, 1201\}$ through a separate patched-fplll campaign whose seeds land in their own tree, never overwriting the canonical seeds.

4 The SD-BKZ Advantage Has a Dimension-Onset on NTRU

On circulant NTRU at $q = 97$ (standard regime), $\beta = 20$, 50 tours, the SD-BKZ-vs-BKZ advantage has a *sharp* dimension transition, not a smooth ramp (20 seeds per n):

Three regimes: tied for $n \leq 67$; an onset zone at $n = 71$ – 73 where the mean spikes to +4–10 nats with $\sim 25\times$ the plateau variance (SD-BKZ cracks the dense sublattice on most-but-not-all instances); and a stable +1.5-nat plateau for $n \geq 79$ (std ≈ 0.2 , 20/20 wins). The onset spike exceeds the plateau—non-monotonic.

5 Fatigue and the Cross-Dimension DSD-Onset Gap

Sweeping q across the fatigue point at fixed n , the reference-free BKZ-vs-SD-BKZ profile divergence peaks near $q \approx q_{\text{fat}}$. The SD-BKZ DSD-onset occurs at progressively *lower* q than the BKZ onset as n grows—the gap widens with dimension:

$n = 127$ is excluded (fplll §8 catastrophic cancellation + an off-grid crack).

6 Cross-Engine: SD-BKZ on a Sieve Oracle

RHF blindness is engine-independent but regime-dependent. On NTRU at $q = 97$, $\beta = 40$, the companion paper’s thesis reproduces on the sieve: in the standard

Table 2: Reference-free DSD-onset: SD-BKZ reaches dense-sublattice discovery at lower q than BKZ, gap growing with n (~ 15 – 20 seeds/cell).

n	SD onset q	BKZ onset q	gap%
67	146	149	2
79	175	175	0
89	237	281	18
101	426	514	21
113	732	932	27

Table 3: Matched cross-engine DSD-onset: SD-BKZ reaches DSD before BKZ on *both* engines ($n = 89$, $\beta = 40$, 50 tours, 20 seeds/cell, $b_1 > 1.5$).

q	fpLLL (enum)		G6K (sieve)	
	SD/BKZ	gap	SD/BKZ	gap
97, 113	0/0	0 (pre-onset)	0/0	0 (pre-onset)
137	8/1	+7	14/4	+10
157	20/18	+2	20/19	+1
≥ 181	20/20	0 (sat.)	20/20	0 (sat.)

regime the RHF of BKZ and SD-BKZ bases is identical to machine precision while $d(\text{LN})$ differs. The effect is regime-dependent—RHF-blindness breaks only on the rare seeds where SD-BKZ finds a short vector (which RHF reflects). This is consistent with RHF-equality reported under a non-exact (approximate-enumeration) oracle [Row24]; the present work measures the structure RHF misses.

Cross-engine validation of the construction. A dimension-resolved fpLLL-vs-G6K comparison ($n = 89/101/113$, 20 seeds each) confirms the self-dual G6K construction is sound: at $n = 101/113$ the two engines agree perfectly (no spurious sieve firing), and where a short vector genuinely exists ($n = 89$, seed 19) both engines land the identical first GS log-norm.

The DSD-onset gap reproduces on *both* engines, confound-free. We run the *same* grid ($n = 89$, $\beta = 40$, 50 tours, $q \in \{97, 113, 137, 157, 181, 211\}$, 20 seeds/cell) on both the fpLLL enumeration engine and the G6K sieve, under the *identical* reference-free DSD criterion: $b_1 = \min_i \|\mathbf{b}_i^*\| > 1.5$ (in log-norm; the q -independent ternary-secret floor $\approx \sqrt{2n} \cdot 2/3 \approx 1.96$). On both engines SD-BKZ reaches dense-sublattice discovery at *lower* q than BKZ.

At $q = 137$ SD-BKZ cracks DSD where BKZ does not—8/20 vs 1/20 on fpLLL, 14/20 vs 4/20 on G6K—the same onset ($q \approx 137$), same direction, same magnitude scale (+7 enum, +10 sieve). This is the DSD-onset gap of Table 2 reproduced across a fundamentally different oracle, and on a matched β : it is therefore *not* a $\beta = 20$ artifact (both engines run $\beta = 40$) and *not* an engine artifact (enumeration and sieving alike). Both onsets sit at lower q than the $\beta = 20$ fpLLL onset ($q=237$)—stronger reduction lowers the onset.

Crucially the gap is visible *only* reference-free: on the seeds where SD-BKZ cracks DSD, the *signed* $d(\text{LN})$ advantage is large and *negative*—mean -21.9 (fpLLL) and -30.8 (G6K) at $q = 137$ —because the DSD Z-shaped profile is far from the LWE-derived $d(\text{LN})$ reference. This makes the “signed $d(\text{LN})$ ” a reference artifact on NTRU” point concrete and cross-engine: report reference-free DSD, never the signed advantage.

7 Re-Validating the fplll GSO Fix on a New Modulus Family

The companion paper documents a catastrophic-cancellation bug in fplll’s squared-form GSO update at cryptographic moduli, and a Kahan-compensated fix. We re-validate that fix on a new (NTRU) modulus family: on $q \in \{971, 1087, 1201\}$ at $n = 127$, the four contaminated seeds recover from the precision-floor clamp ($b_1 = -345.388$) into the control band (≈ -0.1); the patched build passes all 15 fplll regression tests. The fix is thus validated on two modulus families (the companion paper’s $q = 3329$ LWE, $38\% \rightarrow 0\%$ degeneracy, and this NTRU set).

8 Discussion

9 Reproducibility

10 Conclusion

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